

Features of the I2C peripheral:

- I<sup>2</sup>C-bus specification and user manual rev. 5 compatibility:
  - Target and controller modes, multicontroller capability
  - Standard-mode (Sm), with a bitrate up to 100 kbit/s
  - Fast-mode (Fm), with a bitrate up to 400 kbit/s
  - Fast-mode Plus (Fm+), with a bitrate up to 1 Mbit/s and extra output drive I/Os
  - 7-bit and 10-bit addressing mode, multiple 7-bit target addresses
  - Programmable setup and hold times
  - Clock stretching
- SMBus specification rev 3.0 compatibility:
  - Hardware PEC (packet error checking) generation and verification with ACK control
  - Command and data acknowledge control
  - Address resolution protocol (ARP) support
  - Host and device support
  - SMBus alert
  - Timeouts and idle condition detection
- PMBus rev 1.3 standard compatibility
- Independent clock: a choice of independent clock sources allowing the I<sup>2</sup>C-bus communication speed to be independent of the PCLK reprogramming
- Wake-up from Stop mode on address match
- Programmable analog and digital noise filters
- 1-byte buffer with DMA capability

**Table 8. I<sup>2</sup>C implementation**

| I <sup>2</sup> C features <sup>(1)</sup>                     | I2C1 | I2C2 |
|--------------------------------------------------------------|------|------|
| Standard mode (up to 100 kbit/s)                             | X    | X    |
| Fast mode (up to 400 kbit/s)                                 | X    | X    |
| Fast Mode Plus (up to 1 Mbit/s) with extra output drive I/Os | X    | X    |
| Programmable analog and digital noise filters                | X    | X    |
| SMBus/PMBus hardware support                                 | X    | -    |
| Independent clock                                            | X    | -    |
| Wakeup from Stop mode on address match                       | X    | -    |

1. X: supported

### 3.18 Universal synchronous/asynchronous receiver transmitter (USART)

The devices embed four universal synchronous/asynchronous receivers/transmitters that communicate at speeds of up to 6 Mbit/s.

They provide hardware management of the CTS, RTS and RS485 DE signals, multiprocessor communication mode, synchronous SPI communication and single-wire half-